

CHAPTER 1

INTRODUCTION



WHAT IS COGNITIVE PSYCHOLOGY?

- The scientific study of how people and animals process information.
 - We do it a lot
 - Mostly smoothly and effortlessly
 - Sometimes it goes wrong
- Cognitive psychology seeks to answer that broad question and is the study of how humans (and other animals)
 - acquire information
 - store information in memory
 - retrieve information
 - work with information to reach goals.
- Mental representations
 - Image or verbal concept of some external reality
 - Essential in cognitive psychology and topic of much debate

HISTORY AND APPROACHES

- Greek philosophers: Plato and Aristotle
- Memory is a wax tablet:
 - Seems rather passive:
 - experience is simply imprinted on humans in an automatic fashion
- Memory is like a aviary:
 - *"Let us suppose that every mind contains an aviary stocked with birds of every sort, some in flocks apart from the rest, some in small groups, and some solitary and flying in any direction among them all When we are babies we must suppose this receptacle empty, and take the birds to stand for pieces of knowledge. Whenever a person acquires a piece of knowledge and shuts it up in his enclosure, we may say he has learned or discovered the thing of which this is knowledge and this is what 'knowing' means"*
 - Explicit search process for retrieval: *"one is hunting once more for any piece of knowledge that he wants, catching and holding it, and letting it go again"*

HISTORY AND APPROACHES

- Associationism/empiricism
 - John Locke, David Hume, John Stuart Mill, ...
 - all knowledge came from experience
 - associations linked ideas and memories.
- Introspectionism
 - Wilhelm Wundt, Edward Titchener, ...
 - break down complex experiences into elementary sensations.
 - Titchener: *"Just as a chemist would never rely on an untrained assistant for any but the simplest measurements, so also the laboratory psychologist cannot rely on untrained introspectors for any but the crudest observations."*

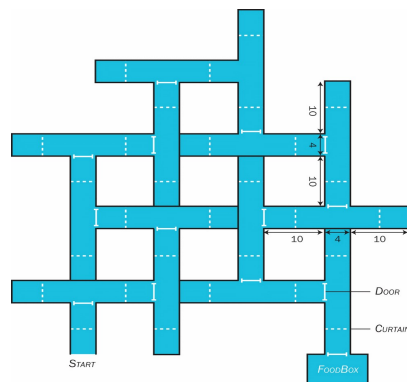
QUOTE FROM TITCHENER

- *If you hold the shaft of a pin loosely between the forefinger and thumb of the right hand, and bring the point down sharply but lightly upon the back of the left hand, you will sense first the impact itself, and then, after a brief but noticeable interval, something finer, like a prick or a thrill. This second sensation is due to a moderate stimulation of the specific organ of pain...*
- *For accurate results, the skin must be shaved, and softened with soap and water; and the surface must be explored by help of a very delicate hair-point. The sensation obtained from the pain spots then occurs in three stages: first, as a bright, itchy sensation; secondly, as a prick or wiry thrill; and thirdly, as punctiform pain. It is always delicate and lively, and has less body than the sensation of pressure.*

BEHAVIOURISM

- John Watson (1913), Edward Thorndike (1898)
- Abandoned the attempt to look inside the mind.
- Only observable behaviour and stimuli as its data.
- Watson:
 - Thinking could be viewed as “covert or inner speech”
 - Cognition was seen as perceptual motor responding.
 - However, curare experiment (Smith et al., 1947):
 - No muscle movement possible, still thinking
- Tolman was less extreme:
 - mental maps are mental representations of a spatial layout
 - latent learning.

TOLMAN'S 14 UNIT T MAZE



- A number of T junctions leading to true path segments or blind alleys which the rat must navigate in order to find the food.
- HR: food reward
- HNR: no food reward
- HNR-R: only food reward after day 11

TOLMAN'S 14 UNIT T MAZE

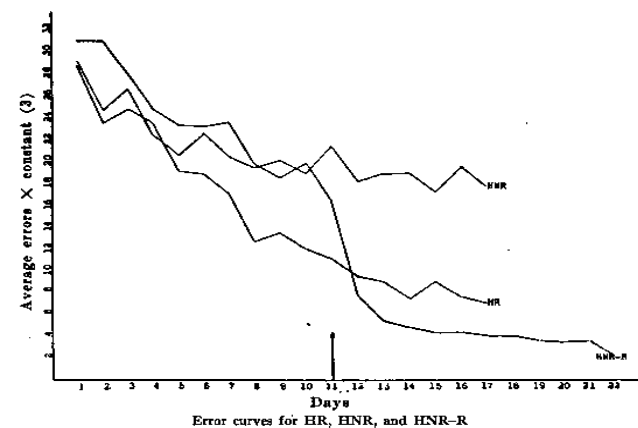


FIG. 6
(From E. C. Tolman and C. H. Honzik, Introduction and removal of reward, and maze performance in rats. *Univ. Calif. Publ. Psychol.*, 1930, 4, No. 19, p. 267.)

- HR: quick learning
- HNR: no decrease in “errors”
- HNR-R: first 11 days cfr HNR, then very quickly cfr HR; hence implicit learning

TOLMAN'S 14 UNIT T MAZE

- An abstract mental representation of the maze had been acquired, a cognitive map
- Question:
 - Did they learn really an abstract map or was it linked to specific motor responses (running)?
- Extra experiment:
 - Same maze, but now filled with water
 - Rats swam to the food immediately
 - Hence, they had acquired an abstract mental representation that was independent of the specific movements that allowed them to complete the maze.

INFORMATION PROCESSING: THE COGNITIVE REVOLUTION

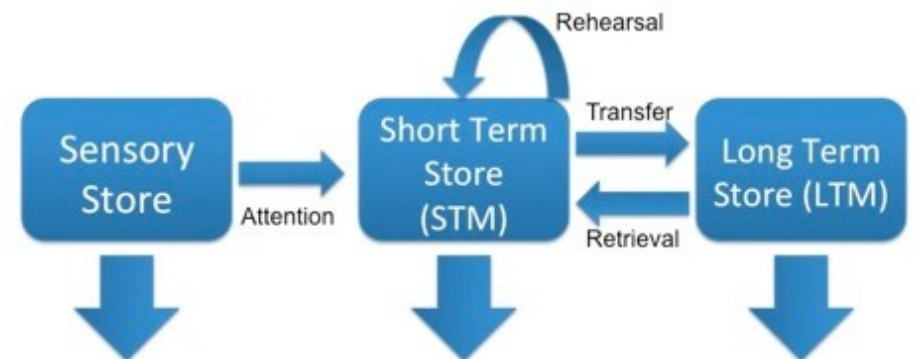
- Important terms for InfoProc-account:
 - Mental operations
 - inner actions manipulating mental representations.
 - Internal representation:
 - Mental representation of an external object or event
 - A substitution of something
 - Working memory/short-term memory
 - short-term limited memory store for holding and manipulating information.
 - Long-term memory
 - virtually unlimited store of information.

INFORMATION PROCESSING: THE COGNITIVE REVOLUTION

- The information processing approach
 - a metaphor for understanding mental activity, based on computing.
- Computer:
 - Could do of course calculations (e.g. trajectories of missiles, working out payrolls, ...)
 - Can it also tackle non-numerical problems, e.g. playing chess, suggesting medical diagnoses, ...?
- Two sorts of programs:
 - A model of human thinking = simulation program
 - A program that seeks to solve the problem as effectively as possible, without the need to mimic human strategies = AI program
 - Nowadays it's a mix (see discussion ToM)

MEMORY

(Atkinson & Shiffrin, 1968)



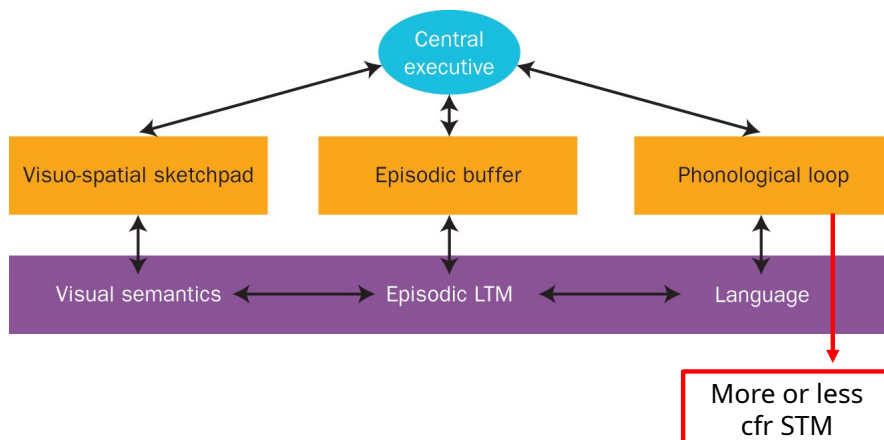
SHORT-TERM MEMORY (ATKINSON & SHIFFRIN, 1968)

- STM
 - holds information in consciousness
 - temporary storage
 - limited capacity
 - used in everyday tasks, language, problem solving
 - most information in STM is not retained.

STM & CHUNKING

- The cognitive system fuses together functionally-related bits of information, a process termed chunking (Miller, 1956).
- A chunk:
 - is when a set of items (e.g., numbers) can be remembered as one (e.g., 12345).
 - enables the accretion of simple units into groups, which can later be used as units themselves.
- Miller estimated that the capacity of short-term memory (memory span) is 7 ± 2 chunks.

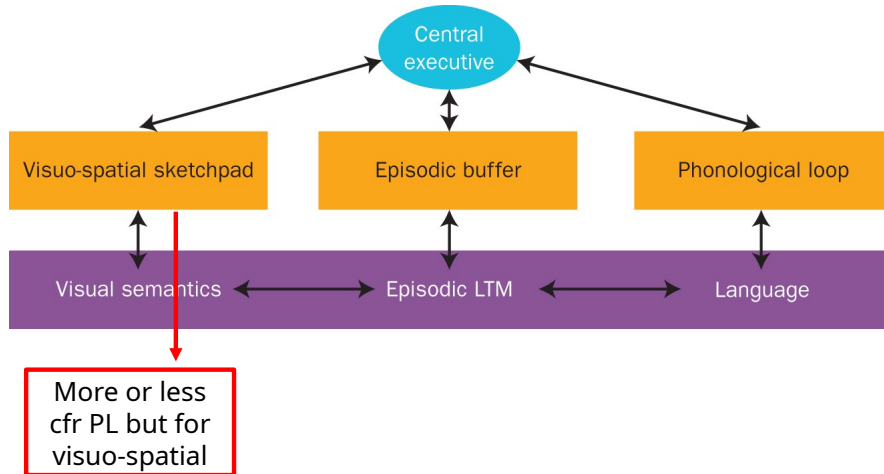
BADDELEY'S WORKING MEMORY MODEL (ALTERNATIVE FOR STM)



THE PHONOLOGICAL LOOP

- Dedicated to speech-based information.
- Consists of two subcomponents that cooperate to keep information for further processing.
- The phonological store:
 - Holds the items online in a passive way
 - Items are characterised by an activity level that decreases over time.
 - Information is lost if it is not refreshed.
 - After two seconds on average, the activity of an item is so weak that it cannot be retrieved anymore and is forgotten.
- Rehearsal:
 - Sets items' activity back to its maximum, item by item.
 - If rehearsal not possible (e.g., double task like de de de) then bad performance

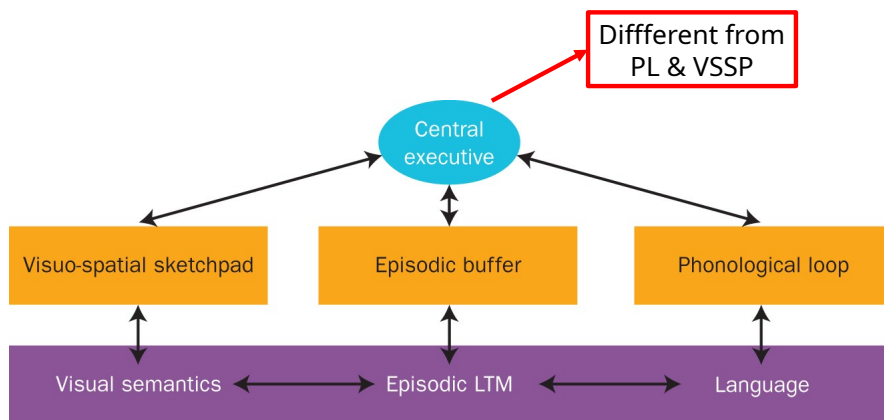
BADDELEY'S WORKING MEMORY MODEL (ALTERNATIVE FOR STM)



THE VISUO-SPATIAL SKETCHPAD

- Holds and manipulates visuospatial information.
- Only 3 or 4 items can be held simultaneously.
- Items are subject to decay.
- VSSP is divided into two subcomponents (Logie, 1995)
 - the visual cache – holds visuospatial information
 - the inner scribe – carries out cognitive operations on this information (e.g. rotating an object).

BADDELEY'S WORKING MEMORY MODEL (ALTERNATIVE FOR STM)



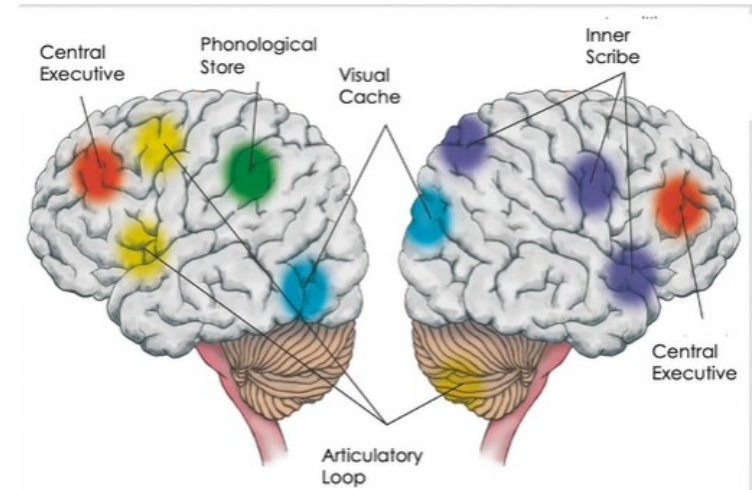
THE CENTRAL EXECUTIVE

- Close to Norman and Shallice's supervisory activating system
- Monitors attentional processes
 - dividing attention, switching attention, and focusing attention on task-relevant items.
 - Link with executive functions
- Controls the information flow within and between the phonological loop and the visuospatial sketchpad.
- Controls the information arriving from the senses and supervises retrieval from memory.
- This component has not been the focus of much research:
 - perhaps due to the fact that it is difficult to pinpoint its specific properties.

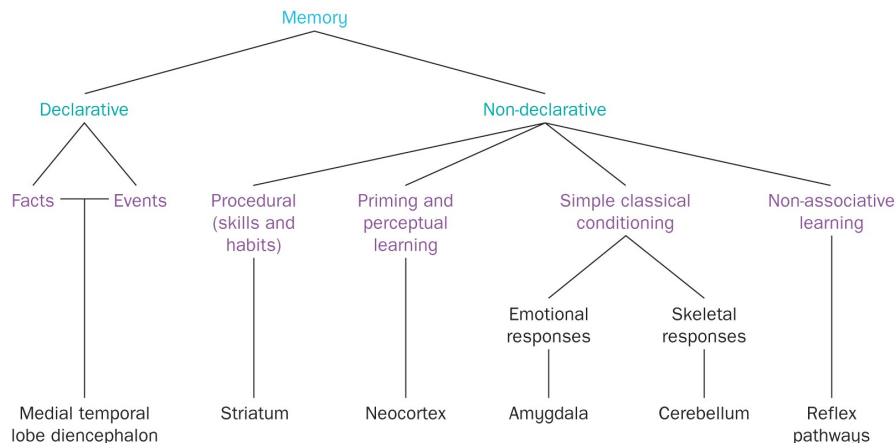
EVALUATION OF BADDELEY'S THEORY

- It does better job than Atkinson and Shiffrin's model at accounting for the data on short-term memory.
- It offers a useful framework for organising a large amount of empirical data on working memory.
- However, it has weaknesses
 - It is rather vague on many details

EVALUATION OF BADDELEY'S THEORY



SQUIRE'S TAXONOMY OF LONG-TERM MEMORY



COGNITIVE CAPACITY/ARCHITECTURE

- For each mental operation, 2 kinds of resources:
 - Knowledge: What we know, have been told, have inferred, ...
 - Cognitive capacity: capabilities that our brain affords us
 - Much of this capacity is probably innate
 - Some of it may be acquired through maturation, practice, ...
- The combination of what we know and what our capacities are is what determines what we do.
- How can we know what is caused by what we know and what is caused by what we can?

A LANGUAGE EXAMPLE AND COGNITIVE PENETRABILITY

- To whom does the pronoun refer:
 1. *John gave the book to Fred because he finished reading it.*
 2. *John gave the book to Fred because he wanted to read it.*
- Pronoun refers to different people:
 - Knowledge of what books are for
 - Knowledge of where things end up when given
- Reference assignment could be easily changed by altering context (encouragement/blind)

A LANGUAGE EXAMPLE AND COGNITIVE PENETRABILITY

- It is within our cognitive capacity
 - to assign a different referent to the pronoun
 - and we can change it
 - in terms of the same principles (Inferences from background knowledge)
- Difference between the cases would be attributed to:
 - a difference in state of knowledge or belief
 - not to a difference in cognitive capacity
 - = cognitive penetrability

BASIC IDEA OF COGNITIVE PENETRABILITY

- If manipulation of beliefs/goals/knowledge systematically alters the empirical phenomenon
 - then the phenomenon is not describing functional architecture and is cognitively penetrable.
- When a pattern is cognitively penetrable
 - then it allows us to conclude that it does not arise from a property of the cognitive architecture

BASIC IDEA OF COGNITIVE PENETRABILITY

- Many/almost any regularity can be systematically altered in a quasi-rational way by imparting new information
 - Hence, many/ most cognitive processes have the characteristic of cognitive penetrability
- Inspection of our conscious experience is not reliable/possible
 - Pronoun assignment:
 - No awareness whatsoever of their being inferences
 - Nevertheless many processes involved

WHAT IS A MENTAL REPRESENTATION ?

- Critical part of psychological explanation
- Source of great confusion
 - (Of course) no clear answer in this course
- Common components of mental representation in cognitive science:
 1. A represented world
 2. A representing world
 3. Representing rules
 4. A process that uses the representation

A REPRESENTED WORLD

- = domain that the representations are about
 - World outside the cognitive system
 - Some other set of representations inside the system

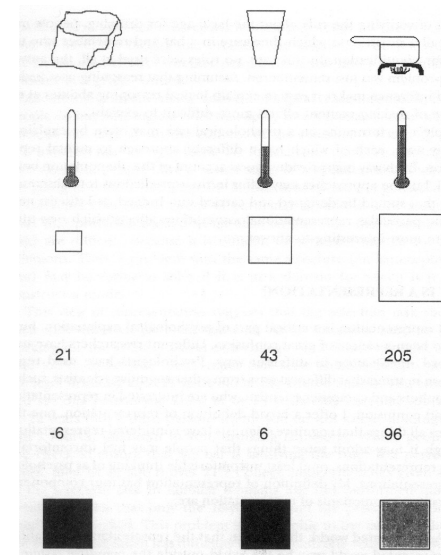
A REPRESENTING WORLD

- = the domain that contains the representations

VARIOUS WAYS OF REPRESENTING INFORMATION

Temperature:

Frozen, room temperature,
boiling



IMPORTANT NUANCE

- Only focus on temperature of water
 - All the rest of info is lost (shape, size, curvature, ...)
- In all known representational systems, the representing world loses information about the represented world.

REPRESENTING RULES

- = the representing world is related to the represented world through a set of rules
 - they map elements of represented world (e.g. temperature)
 - to elements in representing world (e.g. height of mercury)
 - Each temperature is reflected by a unique height of mercury
 - Therefore, an isomorphism between
 - Temperature in represented world
 - Height of mercury in representing world
 - Sometimes homomorphism:
 - Celcius temperature scale with accuracy to only 1 degree as representing world: $20.5^{\circ} = 21.4^{\circ} = 21^{\circ}$
 - The representation has lost information

A PROCESS THAT USES THE REPRESENTATION

- = it makes no sense to talk about representations in the absence of processes
- First 3 components merely create the potential for representation
- Only when there is a process that uses the representation, the system actually represents

TWO SORTS OF REPRESENTATIONS

- The length of the line and area of rectangles:
 - Depict a continuous quantity with another continuous quantity
 - These representations are analog:
 - They have the same inherent structure
 - If line $A > B$ and $B > C$, then $A > C$
 - Temperature has same transitive structure

TWO SORTS OF REPRESENTATIONS

- Not all representations are analog
 - E.g., numerical representations of temperature
- Very different from previous:
 - Making numbers smaller/bigger does not signal change in temperature
 - 20 ₂₀ 20
 - Only change in the digits change the representation
 - Nothing inherent in scratches of ink requires 21 to be larger than 20
 - There is a system of representing rules which links the numerals to abstract mathematical quantities
- These representations are often called symbolic

COMPUTER SIMULATION

- General Problem Solver
 - One of the first simulation models
 - Could tackle problems such as logic tasks and simple games
 - Similar to humans:
 - By breaking problem down in goals and subgoals
 - = means-ends analysis

DAN WILLIAMS (2018)

- Mental representations are
 - structures within the brain
 - that function like familiar external representations such as maps, models, diagrams, pictures, sentences, and so on.
- Will come back to this “definition” in chapter on reasoning (mental models)

MEANS-ENDS ANALYSIS

- *You are in Leuven, you want to visit your friend in Turin.*
- Comparison of goal with starting point, thinking of possible ways of overcoming the difference, choosing the best one
- The selected option may have prerequisite conditions.
 - If they are not met, subgoal created
- Through creation of subgoals, task is broken down into manageable steps.
 - Different means of transportation, plane is the choice, how to airport, how from airport,

MEANS-ENDS ANALYSIS

- Famous example is the following (crypt) arithmetic problem

- DONALD
- GERALD +
- ROBERT
- D=5

- Step 1: If $D = 5$, then $T = 0$
- Step 2: If $O + E = O$, then $E = 9$

- DONALD 5ONAL5
- GERALD + G9RAL5
- ROBERT ROB9R0

- Step 3: If $L + L + 1 = R$, then R is odd (addition of two digits always leads to an even number)
- Step 4: We know that R is odd and bigger than 5, because $5 + G = R$. $E = 9$, hence R has to be 7

- 5ONAL5 5ONAL5
- G9RAL5 + G97AL5 +
- ROB9R0 7OB970

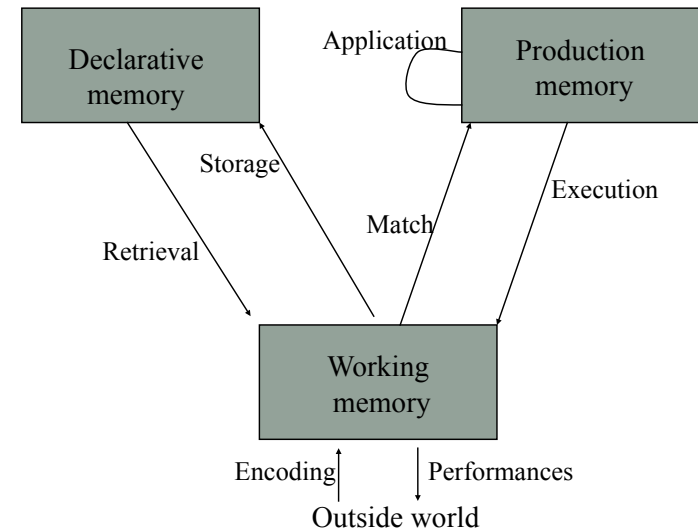
- Step 5: G has to be 1, because $5 + 1 + G = 7$ and G cannot be 0 ($T = 0$).
- Step 6: $L + L + 1 = 7$. Hence, L is 3 or 8. We know that $A + A = 9$, so A is 4, so there is 1 carry-over, so $L = 8$.
- Step 7: $N + 7 = B$, the only digits left are 2, 3 and 6. Hence, $N = 6$ and $B = 3$
- Step 8: Hence, $O = 2$

- 5ONAL5 526485
- G97AL5 + 197485
- 7OB970 723970

COMPUTER SIMULATION

- ACT-R (Anderson, 2004):
 - Series of model; ACT-R is the ultimate successor of a series of increasingly precise models of human cognition
 - HAM, ACT, ACT*, ACT-R (different versions)

SCHEMATIC DIAGRAM OF ACT-MODEL



DECLARATIVE VS PROCEDURAL KNOWLEDGE

- $5 + 7 = 4X$
- Declarative knowledge:
 - 5, 7, 4 are numbers
 - X is a variable
 - You can always divide both sides of an equation by the same number
 - Therefore, you decide to divide both sides of equation by 4
 - This thinking requires great deal of conscious activity

DECLARATIVE VS PROCEDURAL KNOWLEDGE

- $5 + 7 = 4X$
- Procedural knowledge:
 - IF the variable is multiplied by some number, THEN divide both sides of equation by that number
 - IF-part is met, then the THEN-part will fire
 - Efficient problem solving with reduced conscious experience

SKILL ACQUISITION = KNOWLEDGE COMPILATION

- = move from the use of declarative knowledge to procedural knowledge
- Knowledge compilation has two sub-processes:
 - Proceduralisation
 - composition

PROCEDURALISATION

- Transforms declarative knowledge into production knowledge
- During repeated problem-solving episodes, a particular piece of declarative knowledge will occur repeatedly in the context of a particular subgoal
- ⇒ a new production rule is created
- ⇒ concurrent reduction in verbalisation
- ⇒ increase in automaticity

COMPOSITION

- Another method of knowledge compilation
- = cutting out unnecessary productions from a sequence of productions
- In the beginning, many false paths
- Composition prunes away the production rules used in false paths

COMPUTER SIMULATION

- Connectionism
 - Simulate basic learning and perceptual phenomena by means of a large network of simple units:
 - input, output and internal (hidden) units
 - Units are connected:
 - Excitatory or inhibitory links
 - Strength of links:
 - Modified through learning rules, such as backwards propagation, in response to errors, in order to obtain the desired output

COMPUTER SIMULATION

- Connectionism
 - More brain-like than ACT-R
 - Not so complex as real brain
 - Inspired by real neurons and brain organization but highly idealized
- Advantages:
 - Can spontaneously generalize beyond information explicitly given to network
 - Retrieve information even when network is damaged (graceful degradation)

COARSE CODING/ DISTRIBUTED REPRESENTATIONS

	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
concept 1	1	1	1	0	0	0
concept 2	1	0	1	1	0	1
concept 3	0	1	0	1	0	1

(activations of units; 0=off 1=on)

Each unit is involved in the representation of multiple items

DIFFERENT WAYS TO REPRESENT INFORMATION WITH CONNECTIONIST NETWORKS: LOCALIST REPRESENTATION

	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
concept 1	1	0	0	0	0	0
concept 2	0	0	0	1	0	0
concept 3	0	1	0	0	0	0

(activations of units; 0=off 1=on)

Each unit represents just one item
 ☞ “grandmother” cells

ADVANTAGE OF DISTRIBUTED REPRESENTATIONS

- Efficiency
 - Solve the combinatorial explosion problem: With n binary units, 2^n different representations possible.
- Damage resistance
 - Even if some units do not work, information is still preserved – because information is distributed across a network, performance degrades gradually as function of damage
 - (aka: robustness, fault-tolerance, graceful degradation)

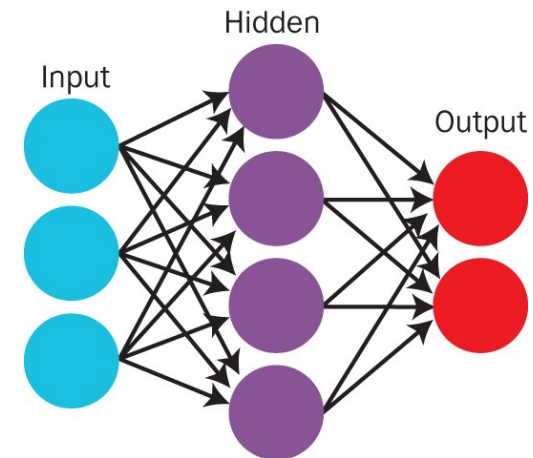
SUPPOSE LOSS OF UNIT 6

	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
concept 1	1	1	1	0	0	0
concept 2	1	0	1	1	0	1
concept 3	0	1	0	1	0	1

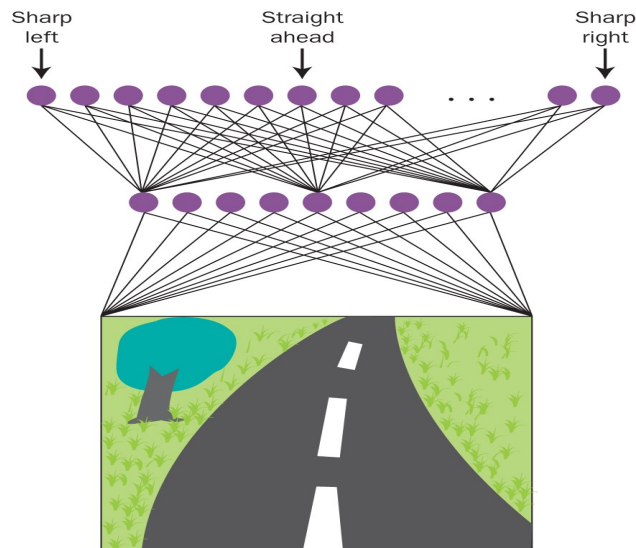
(activations of units; 0=off 1=on)

COMPUTER SIMULATION

- Connectionism
 - Feedforward network:
 - Info flows in one direction from input to hidden to output layers



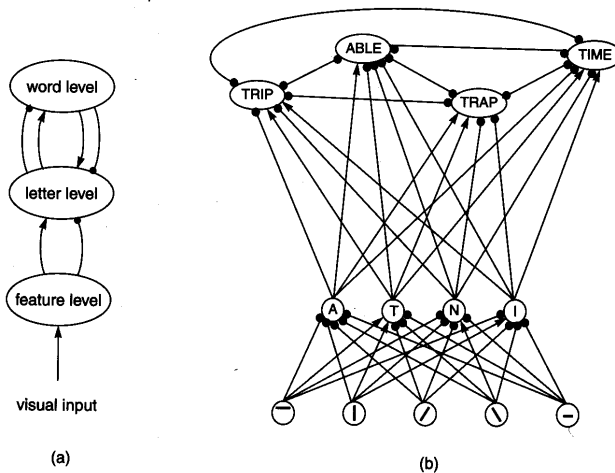
A CONNECTIONIST NETWORK FOR NAVIGATION



COMPUTER SIMULATION

- Connectionism
 - Recurrent network:
 - Some connections feed back to earlier layers
 - E.g., some output units send info back to input layer, in order to take account of previous output when processing next input

MCCLELLAND AND RUMMELHART'S MODEL OF WORD SUPERIORITY EFFECT



SOME TERMS

- Artificial Intelligence
 - Broader term
 - Idea of machines being able to carry out tasks in a way that we would consider "smart"
- True AI:
 - What you see in movies or human level AI,
 - In other words, AI that understands the world in the way we do, interacts with the world like we do

SOME TERMS

- GOFAI (good old-fashioned AI)
 - More classical approach of encoding a model of the problem
 - Expecting the system to process the input data according to this model to provide a solution
 - Need for symbolic techniques is getting a fresh wave of interest of late
 - For AI based systems to be accepted in certain high-risk domains, their behavior needs to be verifiable and explainable
 - The latter is more difficult to achieve by connectionist algorithms

SOME TERMS

- connectionism
 - Set of approaches in the fields of AI, cognitive psychology, cognitive science
 - Modelling mental of behavioral phenomena as the emergent processes of interconnected networks of simple units
 - Term already introduced 1940 (Donald Hebb), first golden age 1980-200
 - Now popular again: Artificial Neural Network, Convolution Neural Networks, Vector Machines, deep learning, machine learning ...
 - Users does not specify the rules of the domain being modelled:
 - Network discovers rules from training data
 - User provides the input data and sample output data
 - Statistical models are applied to adjust the weight coefficient of intermediate variables until best best-fitting model is found

FUNCTIONAL LEVEL OF ANALYSIS

- Information processing approach:
 - Focus on mental software
- Basic question:
 - What strategies are followed in information processing?
 - = question about functions and functional properties
- In principle:
 - Possible to answer without looking to hardware
 - Strong functionalist view:
 - Nature of the brain and the details of underlying neural processes are of no concern for analysis of functional level
- However, modern cognitive scientists are not strong functionalist
- Argumentation
 - Tri-level hypothesis (= Pylyshyn; see also Marr)

THE TRI-LEVEL HYPOTHESIS

- Intelligent systems are organized at 3 distinct levels:
 - The physical or biological level
 - The symbolic or syntactic level
 - The knowledge or semantic level
- This means that different regularities may require appeal to different levels

THE TRI-LEVEL HYPOTHESIS

- A simple example:
Imagine a calculator that is being used to do some basic arithmetic.



How would you explain to someone how the calculator works?

THE TRI-LEVEL HYPOTHESIS

- Explain the physical operations of the calculator (transistors, resistors, capacitors, ...)
- Explain the information processing steps, the program, carried out by the calculator
- Specify the laws of arithmetic, and prove that the behavior of the calculator conforms to these abstract laws

THE TRI-LEVEL HYPOTHESIS

1. Explain the physical operations of the calculator (transistors, resistors, ...)
 - = Biological/physical/implementational level
 - Level at which processes occur in the real world
 - This level studies the actual physical makeup of the information processing system.
 - In the case of humans, this would be the brain - whereas in computers, it would be the hardware.
 - The neural and biological changes that accompany and produce cognitive processes
 - Methods of neuroscience/PDP-models

THE TRI-LEVEL HYPOTHESIS

3. Specify the laws of arithmetic, and prove that the behavior of the calculator conforms to these abstract laws
 - = knowledge/semantic/computational level
 - Overt, meaningful behavior, as described by certain patterns of input and output
 - Methods used include formal proofs using math, logic, etc (linguists, philosophers, computer scientists)

THE TRI-LEVEL HYPOTHESIS

2. Explain the information processing steps, the program, carried out by the calculator
 - = Symbolic/syntactic/algorithmic level
 - The underlying formal, symbolic cognitive processes that produce the overt, meaningful behavior in a given environment
 - Experimental study of behavior (psychologist, psycholinguists, computer scientists)

THE TRI-LEVEL HYPOTHESIS

- Primary level of interest for the CSt
 - = syntactic and semantic level or functionalist level
- Important presumption:
 - Cognitive processes are in an important sense formal and symbolic.
- But, all levels must be taken into account!
- Nowadays:
 - Biological/physical/implementational level becomes more and more important
- In next part:
 - Some key aspects of neuroscience!

SMALL WARNING

- "Therefore, to explain cognition, cognitive scientists not only need to look inside the brain, but also outside. We need to take the structures in the world into account, and not just the physical structures, but also the social and cultural structures. We need to understand that brain in that organism in that particular world. " (Iris Van Rooij, 2018)

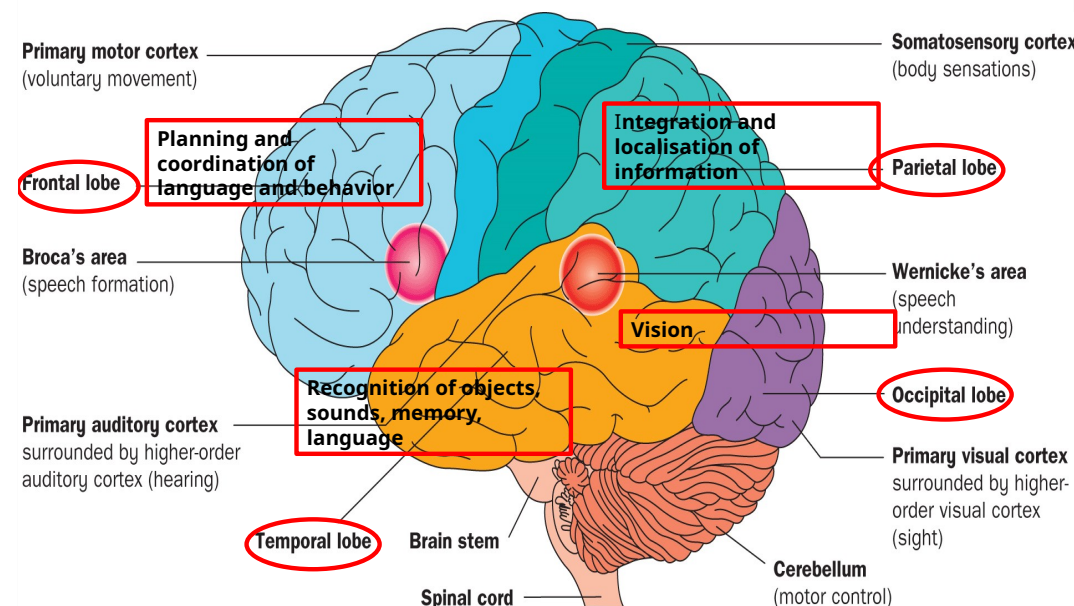
COGNITIVE NEUROSCIENCE

- Cognitive scientists should now stuff about the brain and nervous system
- Brains:
 - controls all reactions of the body and thinking
- Importance of brain:
 - 20% of blood flows to brain, while it represents only 2% of the body
 - Europe: 25% of total cost as a result of disease is caused by disorders in brains (17% by cancer and 5% digestion)

COGNITIVE NEUROSCIENCE

- Brain is central part of nervous system
- It is highly structured into distinct parts and subparts
- Left and right hemisphere (connected by corpus callosum)
- In outer layer of brain (cerebral cortex) four sections:
 - Specialized in processing of different kinds of information

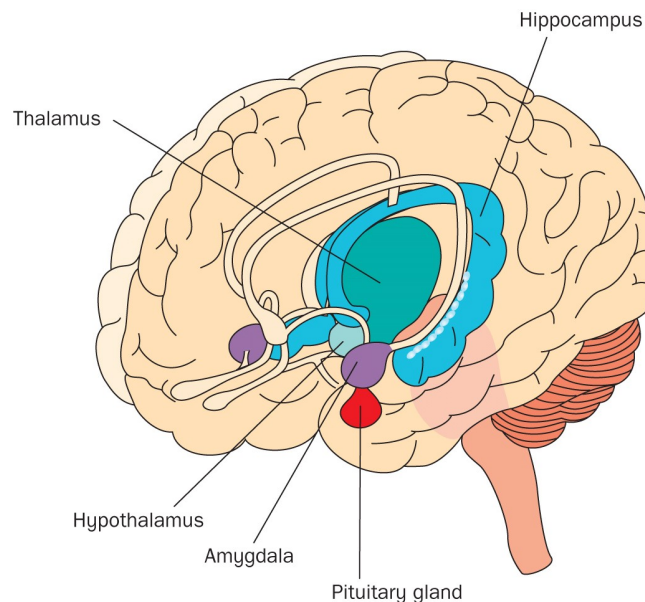
COGNITIVE NEUROSCIENCE



COGNITIVE NEUROSCIENCE

- Deeper in the brain
- Cerebellum (little brain):
 - Important for movement
 - Maybe also supervision of cognitive processes
- Thalamus:
 - Connection Station
 - Control of information flow between the peripheral nervous system and higher layers of the brain
- Hypothalamus:
 - crucial for organization of behaviors related to survival (food, fight, flight, mating)
 - + with pituitary gland: releasing of hormones

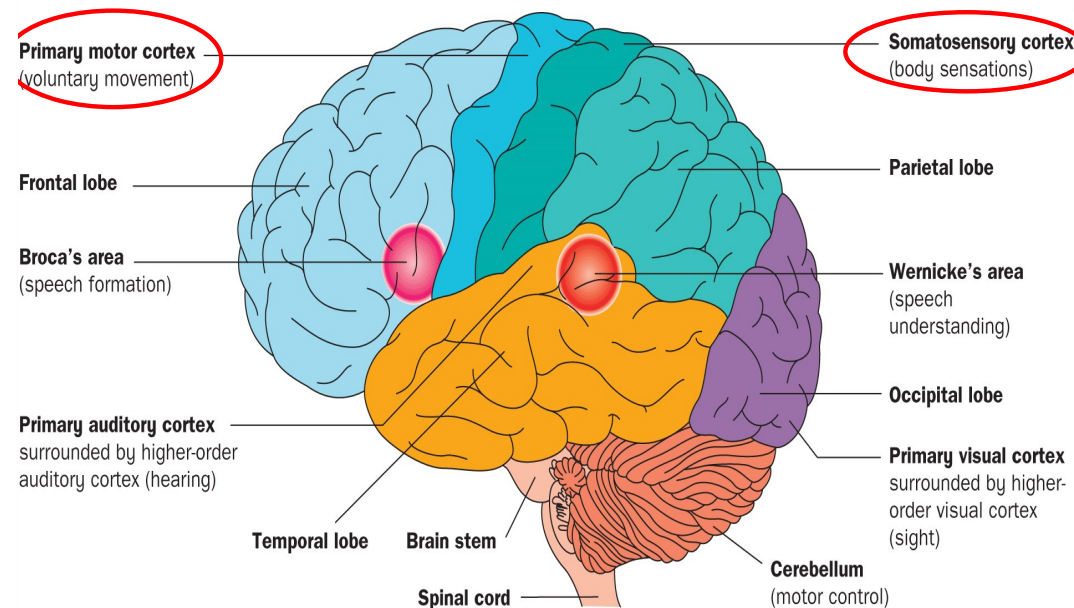
KEY SUB-CORTICAL STRUCTURES OF THE HUMAN BRAIN.



COGNITIVE NEUROSCIENCE

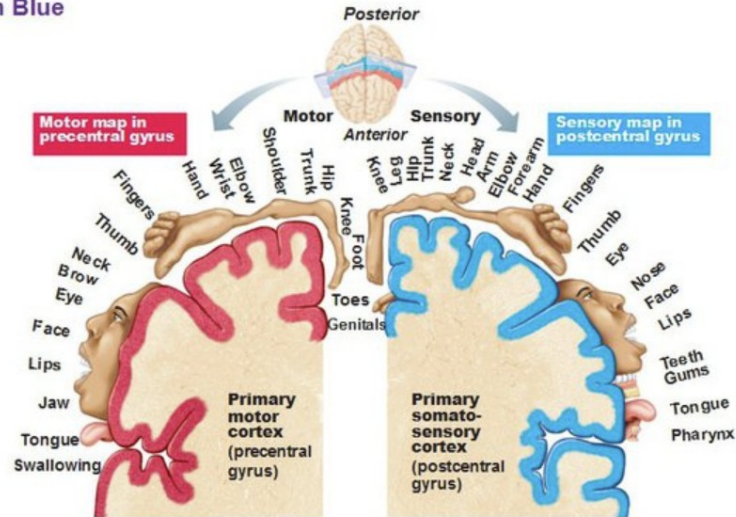
- Deeper in the brain
- Amygdala:
 - primary role in the processing of memory, decision making and especially emotional reactions
- Hippocampus:
 - plays important roles in the consolidation of information from memory and spatial navigation

COGNITIVE NEUROSCIENCE



COGNITIVE NEUROSCIENCE

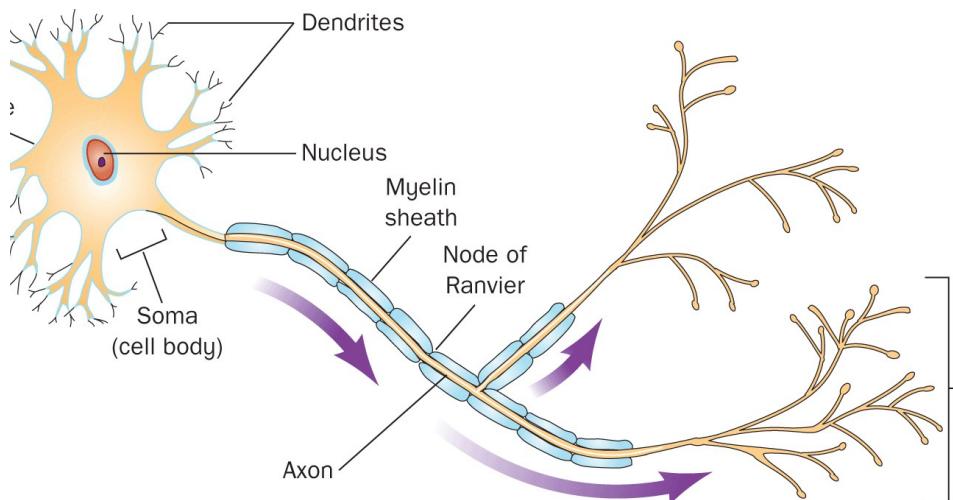
Somatosensory Cortex in Blue



A NEURON

- Building block of brain = neuron
 - Around 86 billion of them
- Basic parts of a neuron:
 - Cell body (soma)
 - Dendrites: they receive signals
 - Axon: transmits signal

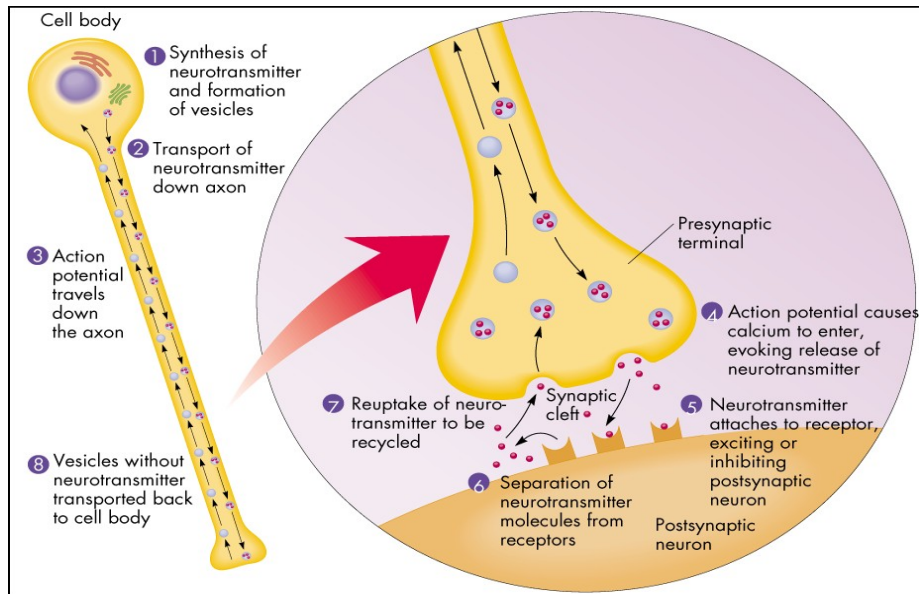
SCHEMATIC OUTLINE OF A NEURON



NEURONAL COMMUNICATION

- Neurons do not touch each other:
 - Synaptic gap
- Communication is chemical transmission:
 - Axon releases neurotransmitters

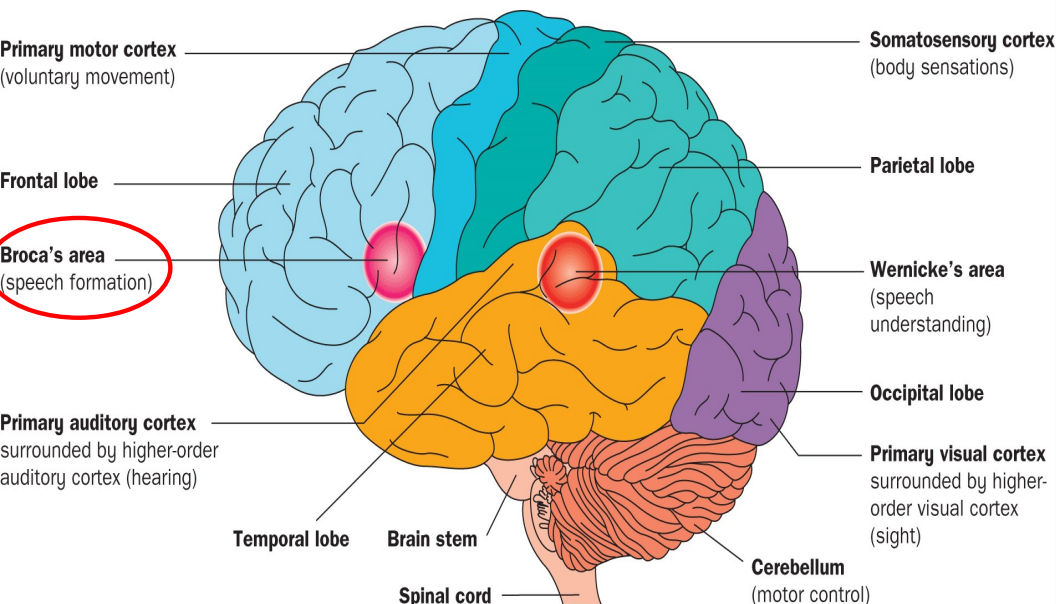
SCHEMATIC OUTLINE OF COMMUNICATION OF A NEURON



COGNITIVE NEUROPSYCHOLOGY

- “What is the effect of a brain damage on behavior?”
- Starting point = Paul Broca, 1861:
 - Patient without normal speech ability after stroke (only “tan”)
 - All other cognitive functions were unaffected
 - Post-mortem examination: small area of damage in inferior part of left frontal gyrus
- = Broca’s area:
 - vital for speech production
- Simple dissociation:
 - Impairment in one function, but not in other function
 - Can be informative with respect to localization

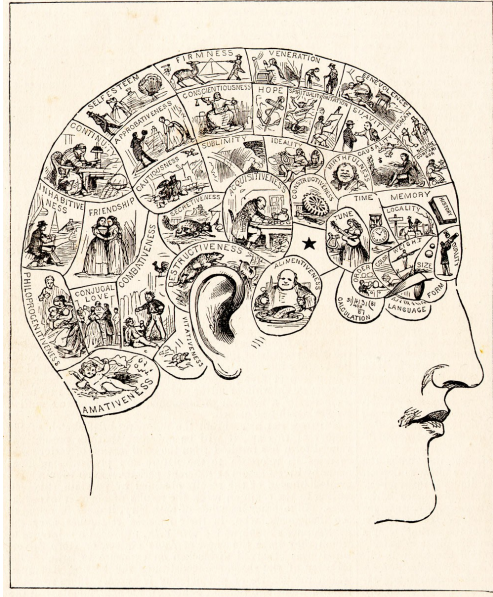
COGNITIVE NEUROSCIENCE



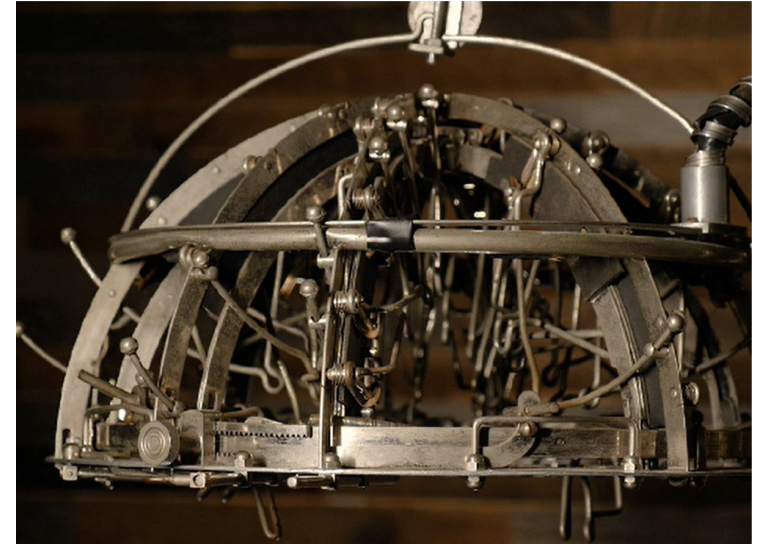
COGNITIVE NEUROPSYCHOLOGY

- Localization
 - Specific mental functions are tied to specific brain areas
 - Also called the modularity hypothesis.
- Early form = phrenology (e.g. Gall)
 - Tried to link psychological functions to brain bumps in the skull
 - Bumps were reflections of growth of brain in specific areas
 - Failed

COGNITIVE NEUROPSYCHOLOGY



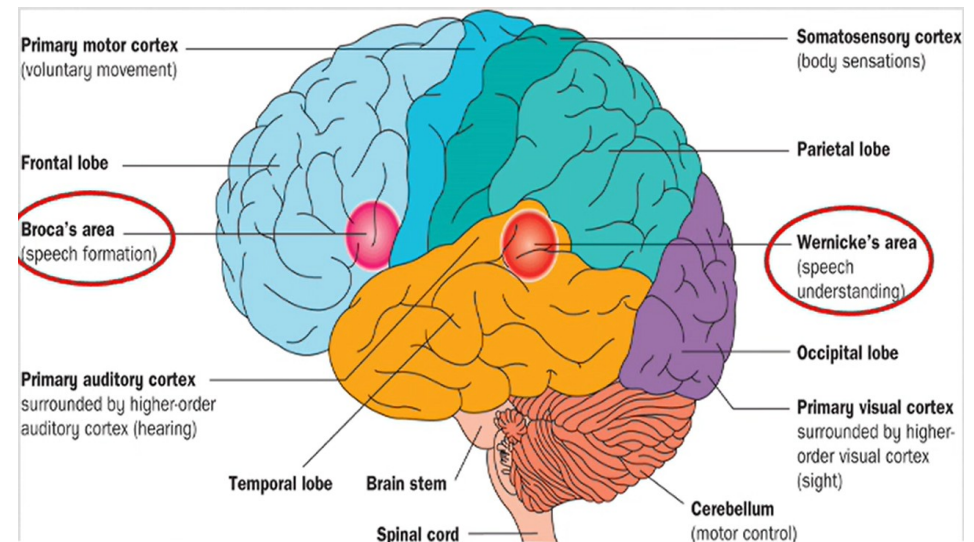
COGNITIVE NEUROPSYCHOLOGY



COGNITIVE NEUROPSYCHOLOGY

- Distributed view:
 - Opposite to modularity
 - Especially complex functions are not localized, but distributed through the brain
 - Quantity of tissue destroyed is more important than the specific site
 - Some support for it, but also for localisation
- Especially interesting = double dissociation:
 - Two patients with different brain injuries
 - Patient A impaired on Function X, ok on Function Y
 - Patient B impaired on Function Y, ok on Function X
 - © Function X and Y are separate functions

COGNITIVE NEUROPSYCHOLOGY



COGNITIVE NEUROPSYCHOLOGY



COGNITIVE NEUROPSYCHOLOGY



WHAT IS A MENTAL MODULE?

- Different theorists have different opinions.
- In common:
 - Mental structures that can be invoked to explain various cognitive capacities.
 - Domain-specific: dedicated to solving very restricted classes of problems
- Two fundamental different sorts of mental structures:
 - Systems of mental representations = Chomskian modules
 - Computational mechanisms = Computational modules

CHOMSKIAN MODULES

- A domain-specific body of mentally represented knowledge/information
- Chomsky: internally represented grammar
- Other examples are e.g. Carey and Spelke:
 - Young children have domain-specific, mentally represented theories for physics, psychology and mathematics

2 CONSTRAINTS

- Not just any domain-specific collection of mental representations = Chomskian module
- 2 important prerequisites
 - Innateness and restrictions on information flow.
 - E.g. Universal Grammar is an innate system of mental representations and most info is not accessible to consciousness

PERIPHERAL VERSUS MASSIVE MODULARITY

- Evolutionary psychology:
 - The mind is largely or even entirely composed of Darwinian modules
- Fodor:
 - famous for his arguments in favour of the modularity of the mind
 - However, argues only in favour of peripheral modularity

COMPUTATIONAL MODULES

- = domain-specific, computational devices
- 1. Cfr classical computers = symbol-manipulating devices (?↔PDP)
- 2. Dedicated to solving problems in a specific domain and only capable of solving problems in this domain
- 3. Relatively autonomous components of the mind: They perform their own internal info processing unperturbed by external systems
- 4. Chomskian modules = systems of representations
Computational models = processing devices (they manipulate representations)
However, both sorts of modules can work together.

ARGUMENTS FOR MASSIVE MODULARITY

- Basic engineering principle:
 - The same machine is rarely capable of solving two different problems equally well.
- Body:
 - Divided into organs for exactly that reason!
 - Pumping blood throughout body ≠ detoxifying poisons
 - Hence, heart ≠ liver
- Likewise, mind consists of large number of circuits that are functionally specialized.
 - These mini-computers are often called modules

ARGUMENTS FOR MASSIVE MODULARITY

- *"human mind is not a general-purpose computer, but a collection of instincts adapted for solving evolutionary significant problems – the mind is a Swiss Army knife"* (Pinker, 1994)



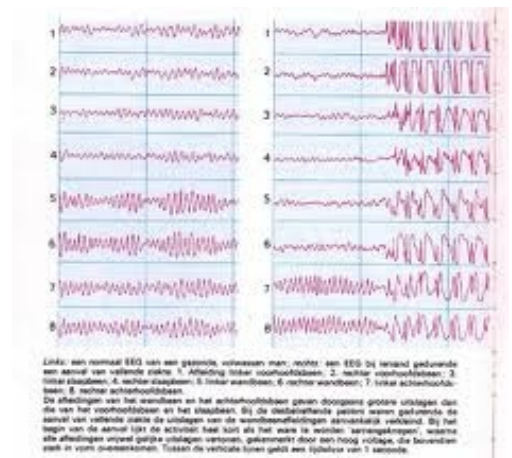
BRAIN IMAGING

- Two main categories:
 - Structural imaging:
 - Shows static anatomy of brain
 - Functional imaging:
 - Represents brain activity over time
- Magnetic resonance imaging (MRI):
 - Dominant method for structural imaging
 - Uses radio waves and a strong magnetic field which surrounds the person scanned
- Electroencephalography (EEG) :
 - Shows waves of electrical activity from scalp recorders
 - Able to measure activity with the precision of milliseconds
 - Localisation of source of activity is in order of centimetres

ARGUMENTS FOR MASSIVE MODULARITY



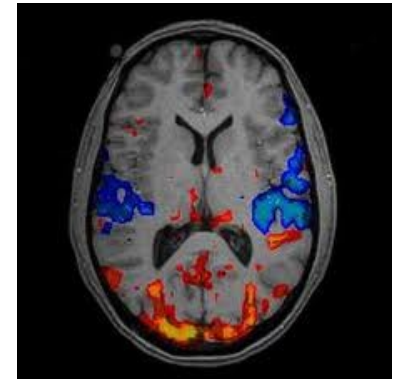
EEG



BRAIN IMAGING

- Positron emission tomography (PET):
 - Uses positron emissions from radioactive glucose to indicate areas of increased blood flow
 - Only shows activity averaged over about 90 sec or longer
 - Hence only crude picture of the sequence of activity
- Functional Magnetic Resonance Imaging (fMRI):
 - Currently favoured technique
 - Uses oxygenation levels of blood flow
 - Good temporal and spatial resolution:
 - Short time scale: 1-3 sec
 - Small areas: 3 cubic mm (voxels)

FMRI



TRANSCRANIAL MAGNETIC STIMULATION (TMS)

